Cisco PE Router KR-PE-15 Standard Operating Procedure for Minor Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) Failure

Node: KR-PE-15 | **Vendor:** Cisco | **Severity:** Minor | **Alarm Code:** ALARM_1555

Description

The Cisco PE router KR-PE-15 at the KR Puram Access site experiences a routing protocol flap failure, characterized by BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This failure domain is precipitated by the dynamic nature of routing protocols, which necessitate continuous neighbor relationships and route exchange. In the context of the PE role on Cisco equipment, the BGP routing protocol, in particular, is prone to flapping due to its reliance on TCP connections and periodic route refreshes. When a BGP peer connection is lost or a route is withdrawn, the PE router initiates a route refresh, which can lead to a BGP route churn. Similarly, OSPF adjacency loss occurs when a neighboring router fails to establish a reliable adjacency, resulting in a loss of routing information. ISIS LSP flooding, on the other hand, occurs when an ISIS LSP is lost or corrupted, leading to an overflow of LSPs in the network. The Cisco PE router's ability to rapidly adapt to these dynamic changes can sometimes result in a routing protocol flap, manifesting as a BGP route churn, OSPF adjacency loss, or ISIS LSP flooding. To mitigate this issue, it is essential to perform a thorough diagnostic analysis and implement targeted resolution steps to ensure the PE router and its neighboring devices operate in a stable and harmonious manner.

Pre-requisite Checks

- Check: Verify the status of the redundant link at KR Puram Access site to ensure it is operational.

- Check: Dispatch remote hands to the KR Puram Access site to verify physical connections and network cabling.

Potential Root Causes

Cause	Probability
BGP route churn caused by periodic route refreshes or lost peer connections.	High
OSPF adjacency loss due to unreliable neighbor relationships or network partitions.	Medium
ISIS LSP flooding resulting from lost or corrupted LSPs.	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version is 1 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State PrepQ Keeper AltTblNat Description 192.168.1.1
```

```
4 65001 1000 1000 1 0 0 00:30:32 ESTABLISHED 0 0 No No, Description:
This command displays the BGP neighbor table, including the state of
each neighbor, the number of messages received and sent, and the table
version.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 192.168.1.1 1 FULL/ - 00:00:38 192.168.1.1
GigabitEthernet0/0, Description: This command displays the OSPF
neighbor table, including the state of each neighbor, the priority,
and the interface used for communication.

Command: show isis database, Expected Output: LSP ID :
192.168.1.1.00-00-0000-0000-0001 Sequence Number : 0x80000002 Checksum
: 0x1234 LSP Length : 32 LSP Type : Level-2 LSP LSP ID :
192.168.1.1.00-00-0000-0000-0001 Sequence Number : 0x80000002 Checksum
: 0x1234 LSP Length : 32 LSP Type : Level-2 LSP, Description: This
command displays the ISIS database, including the LSP ID, sequence
number, checksum, and LSP type.
```

Resolution Steps

Step 1: Step: Verify BGP neighbor relationships, Command: show ip bgp summary, Description: Verify that all BGP neighbors are established and stable.

Step 2: Step: Check OSPF neighbor relationships, Command: show ip ospf neighbor, Description: Verify that all OSPF neighbors are established and stable.

Step 3: Step: Clear ISIS LSPs, Command: clear isis database, Description: Clear the ISIS LSP database to resolve any flooding issues.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 1,
main routing table version is 1 Neighbor V AS MsgRcvd MsgSent TblVer
InQ OutQ Up/Down State PrepQ Keeper AltTblNat Description 192.168.1.1
4 65001 1000 1000 1 0 0 00:30:32 ESTABLISHED 0 0 No No, Description:
Verify that BGP neighbor relationships are stable and established.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 192.168.1.1 1 FULL/ - 00:00:38 192.168.1.1
GigabitEthernet0/0, Description: Verify that OSPF neighbor
relationships are stable and established.

Command: show isis database, Expected Output: LSP ID :
192.168.1.1.00-00-0000-0000-0001 Sequence Number : 0x80000002 Checksum
: 0x1234 LSP Length : 32 LSP Type : Level-2 LSP LSP ID :
192.168.1.1.00-00-0000-0000-0001 Sequence Number : 0x80000002 Checksum
: 0x1234 LSP Length : 32 LSP Type : Level-2 LSP, Description: Verify
that the ISIS LSP database is stable and clear of any flooding issues.
```

Roll-back Procedures

- Step: Restore previous ISIS LSPs, Command: show isis database | append , Description: Restore the previous ISIS LSP database from a saved file.

- Step: Re-establish BGP neighbor relationships, Command: clear ip bgp *, Description: Re-establish BGP neighbor relationships by clearing the BGP table and restarting the BGP process.